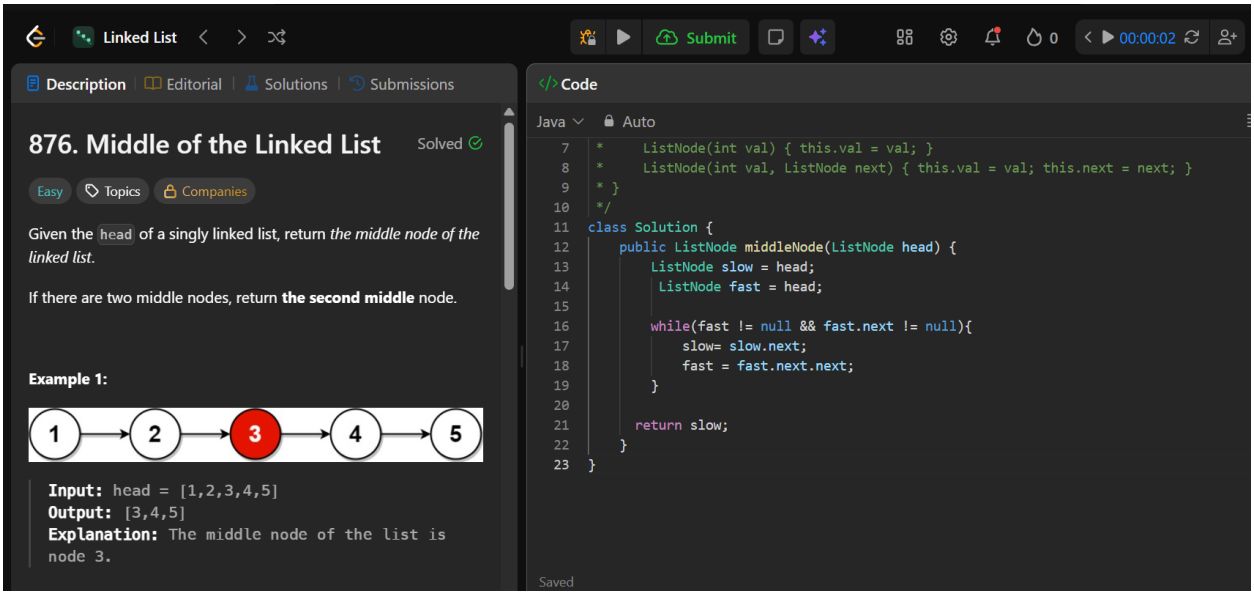
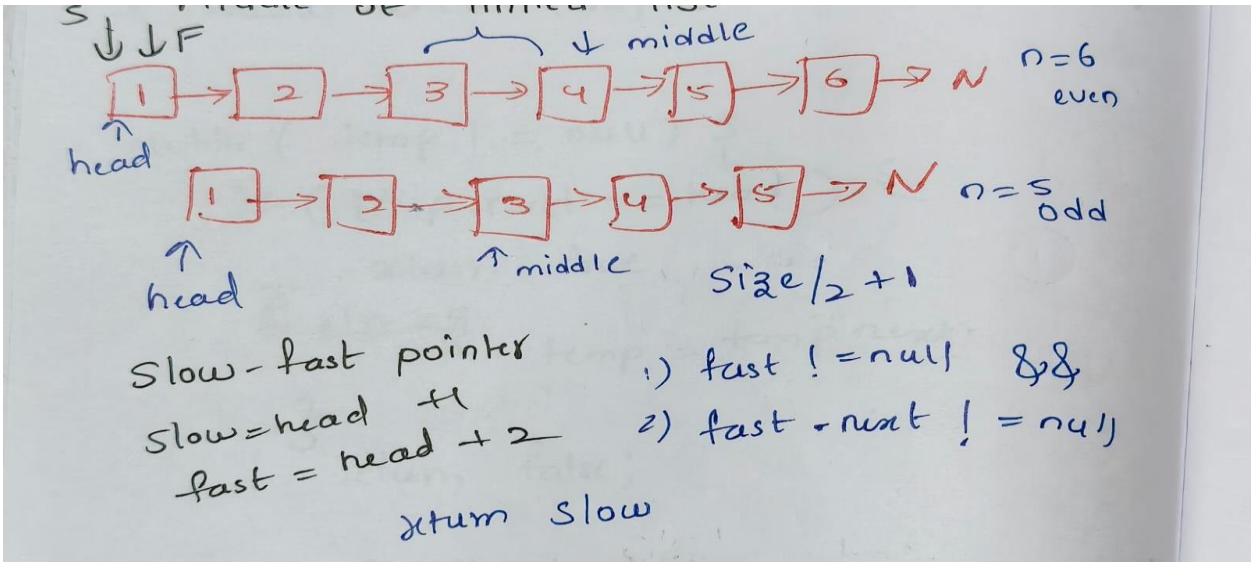
T. DHANA LAXMI
2505A41L46

1.Middle of the linked list LC-876

https://leetcode.com/problems/middle-of-the-linked-list/





2.linked list cycle LC-141

https://leetcode.com/problems/linked-list-cycle/

Description | Editorial | Solutions | Submissions

141. Linked List Cycle

Solved

Easy | Topics | Companies

Given `head`, the head of a linked list, determine if the linked list has a cycle in it.

There is a cycle in a linked list if there is some node in the list that can be reached again by continuously following the `next` pointer. Internally, `pos` is used to denote the index of the node that tail's `next` pointer is connected to. **Note that `pos` is not passed as a parameter**.

Return `true` *if there is a cycle in the linked list*. Otherwise, return `false`.

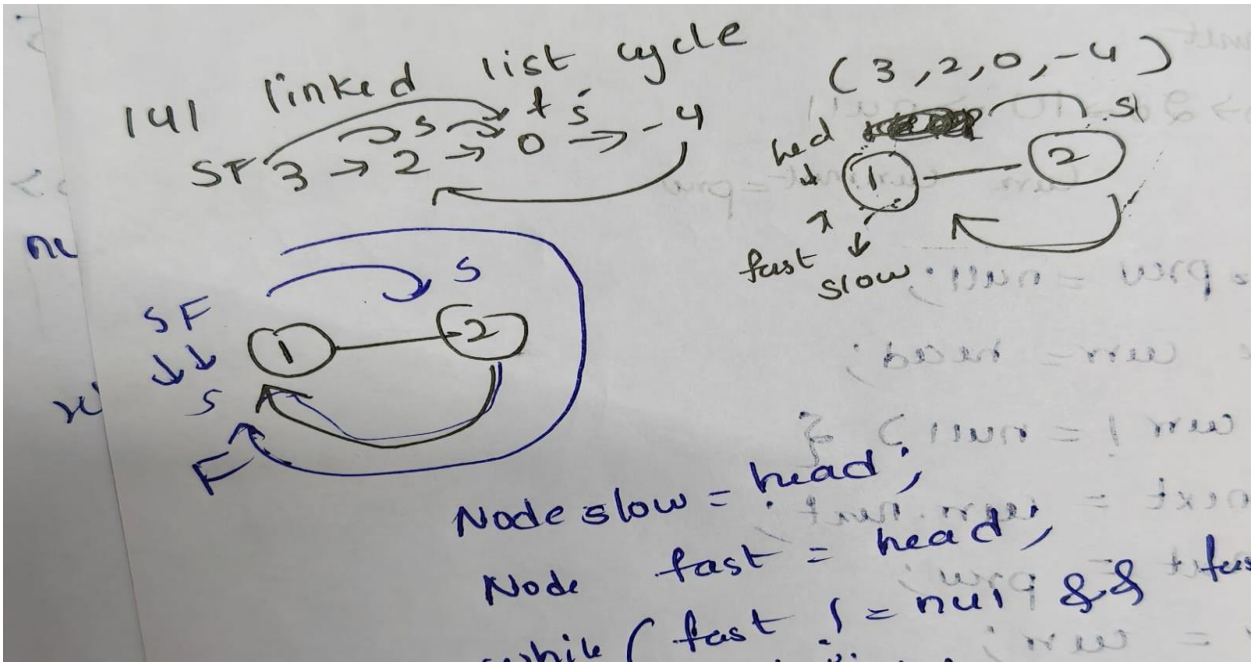
Example 1:

Code

Java | Auto

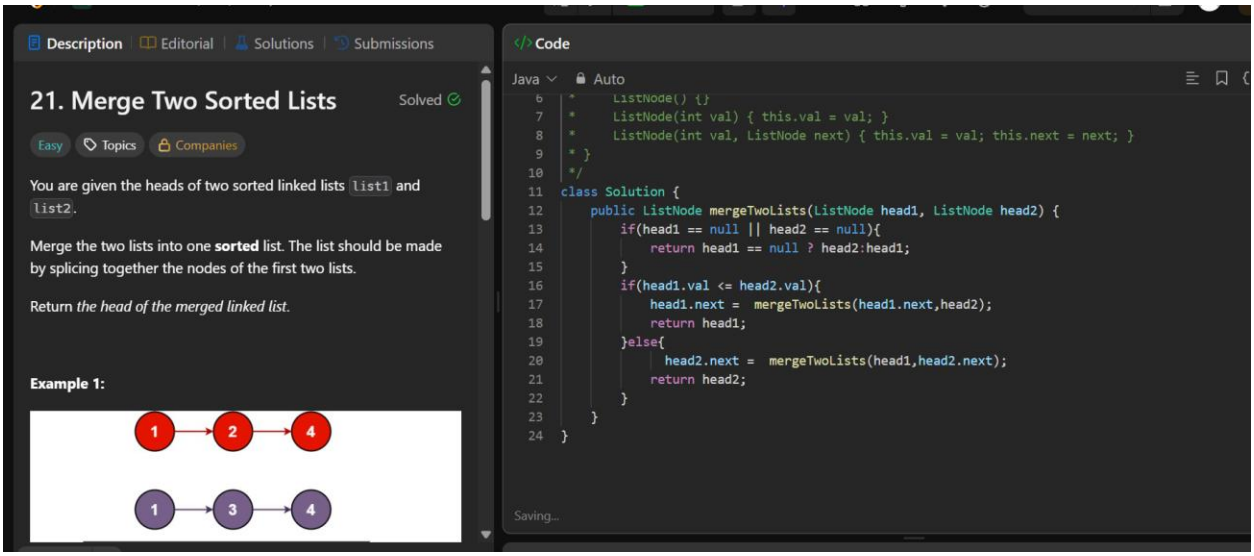
```
*         next = null;
*     }
* }
*/
public class Solution {
    public boolean hasCycle(ListNode head) {
        ListNode slow=head;
        ListNode fast=head;
        while(fast != null && fast.next != null){
            slow = slow.next;
            fast = fast.next.next;
            if(slow == fast){
                return true;
            }

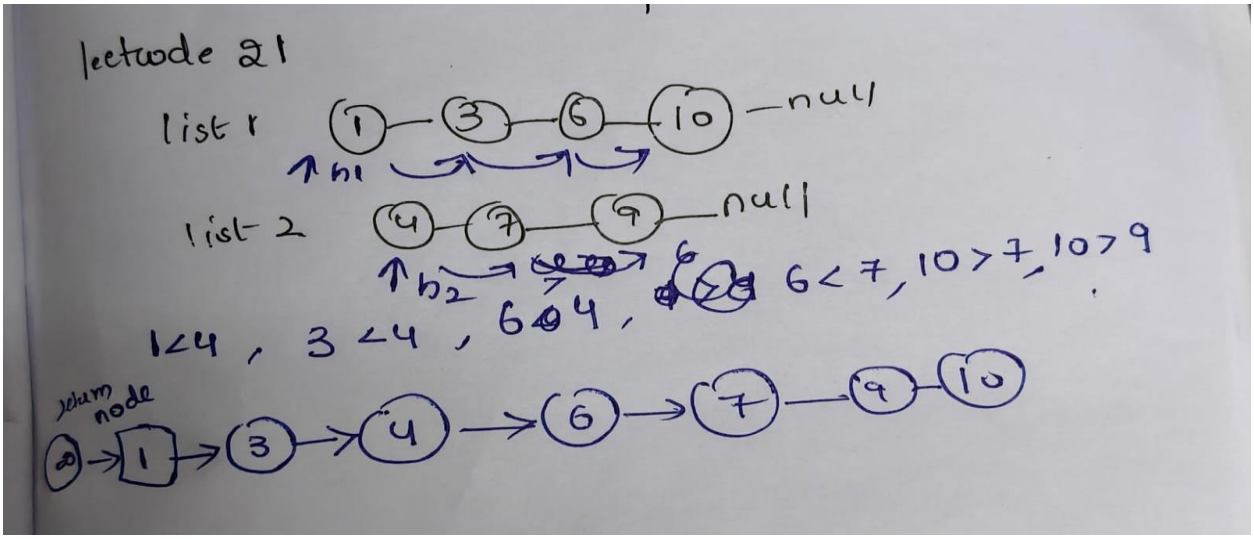
        }
        return false;
    }
}
```



3.Merge two sorted lists

https://leetcode.com/problems/merge-two-sorted-lists





4.reverse linked list

https://leetcode.com/problems/reverse-linked-list

Description | Editorial | Solutions | Submissions

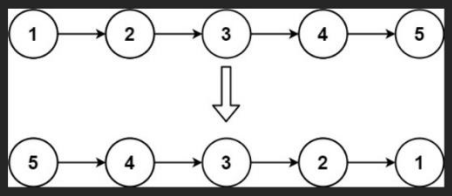
206. Reverse Linked List

Solved

Easy | Topics | Companies

Given the `head` of a singly linked list, reverse the list, and return *the reversed list.*

Example 1:



Input: head = [1,2,3,4,5]
Output: [5,4,3,2,1]

25.1K | 462 | 78 Online

Code

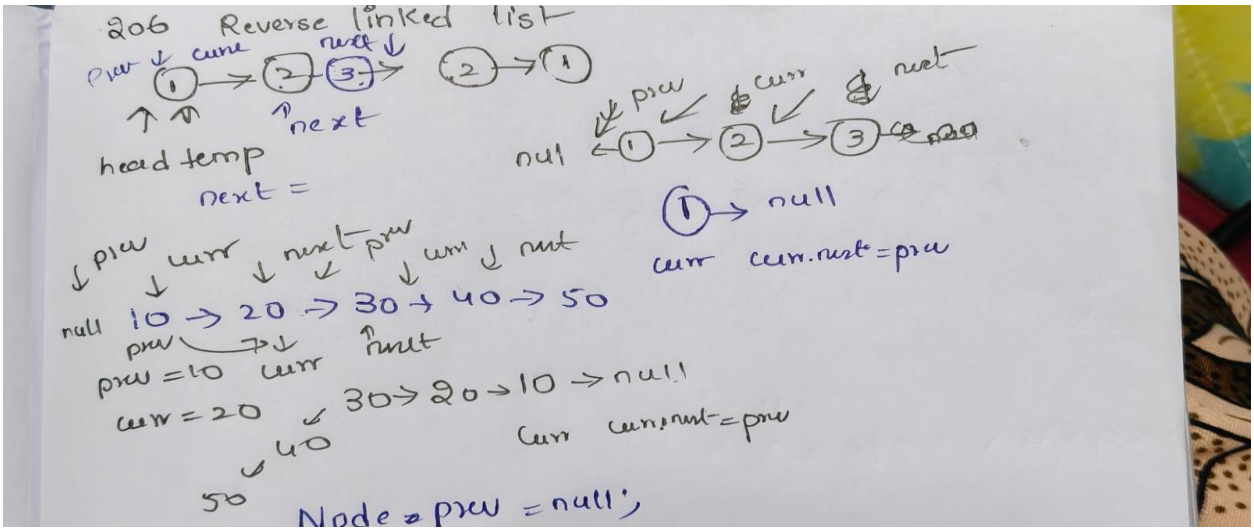
Java | Auto

```
*     ListNode(int val) { this.val = val; }
*     ListNode(int val, ListNode next) { this.val = val; this.next = next; }
* }
*/
class Solution {
    public ListNode reverseList(ListNode head) {

        ListNode prev = null;
        ListNode curr = head;
        while(curr != null){
            ListNode next = curr.next;
            curr.next = prev;
            prev = curr;
            curr = next;
        }
        return prev;
    }
}
```

Saved

Testcase | Test Result



5.remove nth node from end of list

https://leetcode.com/problems/remove-nth-node-from-end-of-list

Linked List

Submit

00:00:02

Description | Accepted | Editorial | Solutic

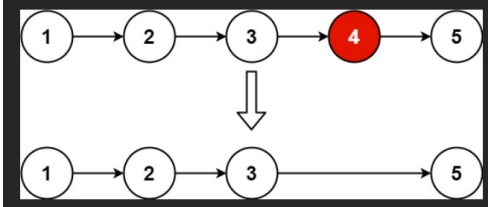
19. Remove Nth Node From End of List

Solved

Medium | Topics | Companies | Hint

Given the `head` of a linked list, remove the `n^th` node from the end of the list and return its head.

Example 1:



21.8K | 406 | 70 Online

Code

Java | Auto

```java
class Solution {
    public ListNode removeNthFromEnd(ListNode head, int n) {
      ListNode dummy = new ListNode(0);
        dummy.next = head;
       ListNode slow = dummy;
        ListNode fast = dummy;

        for(int i = 0; i <= n; i++)  fast = fast.next;
        while(fast != null) {
            slow = slow.next;
            fast = fast.next;
        }
       slow.next = slow.next.next;
        return dummy.next;
    }
}
```

Saved

Testcase | Test Result

